HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

Bachelor Thesis

Modernisation and reliability improvement of the ICNML biometric trace platform

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Academic year	2025-26

Yverdon-les-Bains, 03.03.2026

Authentication

I hereby certify that I have completed this work myself and have used no other sources than those expressly mentioned.

Quentin Surdez

Yverdon-les-Bains, 03.03.2026

Preamble

This Bachelor Thesis (hereinafter BT) is completed at the end of the study program, with a view to obtaining the title of Bachelor of Science HES-SO in Engineering.

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HEIG-VD
The Head of Department TIC

Yverdon-les-Bains, 03.03.2026

Abstract

Bachelor Thesis 2025-26
Title: Modernisation and reliability improvement of the ICNML biometric trace platform
Subtitle:

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Student	**Quentin Surdez**
Supervisor	Prof. Sylvain Pasini
Mandating Company	UNIL - ESC

Specifications

Problem summary

ICNML (*International Close Non-Matches Library*) is an open-souce platform designed for handling biometric traces in scientific and experimentation contexts. It is deployed by several research institutions, including the University of Lausanne (*UNIL*). It is used by collaborators worldwide on projects focused on image analysis and biometric recognition.

Problem statement

The system was initially developed in a research environment that prioritized functionnality and flexibility over long-term maintenance and technical standardisation. Today, the plateform remains a central component in many scientific activities. However, its evolution and sustainability are limited by accumulated technical debt. These weaknesses affect maintenability, security, and usability for both admins and end users alike.

Specifications

Objectives

This aim of this bachelor thesis is to strenghten the security and maintenability of the **ICNML** platform. The work is built around the 3 following objectives.
The first objetive is to **reinforce technical maintainability** by producing documentation of the system architecture and all critical processes. The underlying objective is to enable future contributors to understand, install, and develop the platform easily compared to now.
The second objective is to **improve security** by bringin transparency to internal processes:

- data encryption in the db
- access control mechanisms
- watermark on the images downloaded
- encryption mechanisms applied per donor

The third objective is to **demistify and reinforce management operations** such as installation and backup restoration. This will improve maintanability and reusability by new developers.

The third objective is to **optimise user experience** through filtering and management features. Especially filtering by donor or image quality.

Schedule

This work begins the 16.02.2026 and ends the 23.07.2026. Over the 16 first weeks, from 16.02 to 08.06, the workload is 12 hours per week. The 6 last weeks, from 08.06 to 27.07, the workload is the equivalent of a full-time.
An assessed intermediary submission is due on 20.05. The final report is due on the 27.07 at 12h00.
The defense will be organized between the 24.09 and the 11.10.

Tasks

The work is built into seven phases, each one building ontop of the previous one.
The **Starting phase** (16-19.02) covers the creation of an initial planning document and the drafting of this specification.
The **First steps in the codebase phase** (24.02-02.04) involves an in-depth analysis of the current codebase, the production of initial general technical documentation, and the setup of a **reproducible** local dev environment.
The **Watermark + backup management phase** (07.04-23-04) is focus on the database encryption mechanisms as well as the process behind the backup mechanism. It will also analyse the process of watermarking the images when they are downloaded from the webapp.
The **User management phase** (23.04-07.05) analyses how the keys for the users are generated and stored, how their biometric data are encrypted and how the auth is managed within the application.
The **Deployment management phase** (07.05-20.05) analyses the current deployment solution and produces corresponding documentation. By the end of this phase, the created documentation will give enough information to take a decision on which functional improvements will be implemented during the development phase.
The **Development phase** (20.05-10.07) covers the implementation of the improvements selected. The exact scope is to be determined based on the findings of the precedent phases.
The **Admin end of the project phase** (13.07-23.07) will focus on writing the final report and the different admin tasks that comes with the end of this bachelor thesis such as a publishable summary and a poster.

Deliverables

Here are the expected deliverables :
1. Technical documentation on:
   - System architecture
   - Per-donor image encryption process with key generation and storage
   - Digital watermarking process applied to downloaded images

- Backup mechanism
- Deployment procedure

2. Functional improvements:
   - Scope defined on 20.05, once documentation is complete
   - Candidate features are as followed:
     - Sorting by donor
     - Sorting by image quality
3. End of thesis deliverables:
   - An intermediary report due on 20.05
   - A final report due on 23.07
   - A publishable summary and a poster

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1 Introduction

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2 Planning

2.1 Initial planning

This is the initial planning for the bachelor thesis. It explains the different tasks needed to complete the project. The tasks have their estimated hours needed to complete next to them in parentheses (x). Some tasks have greater estimated hours than their complete subtasks, it is because the hours estimated to write the reports are within the total estimate. \
It is important to know the work begins the 16.02.2026 and ends the 23.07.2026. Over the 16 first weeks, from 16.02 to 08.06, the workload is 12 hours per week. The 6 last weeks, from 08.06 to 27.07, the workload is the equivalent of a full-time.

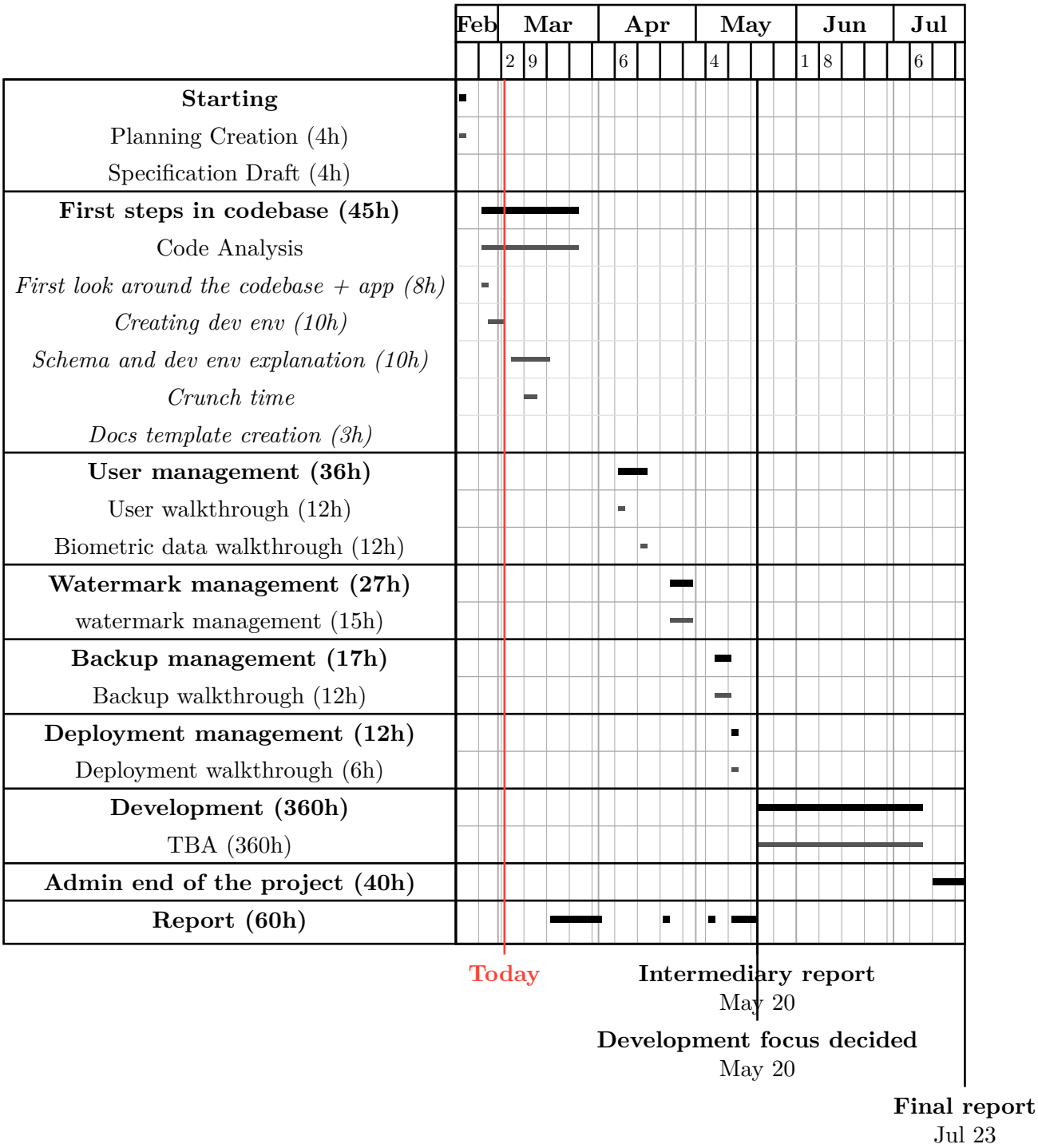
Feb
Mar
Apr
May
Jun
Jul
2
9
6
4
1
8
6
Starting
Planning Creation (4h)
Specification Draft (4h)
First steps in codebase (45h)
Code Analysis
First look around the codebase + app (8h)
Creating dev env (10h)
Schema and dev env explanation (10h)
Crunch time
Docs template creation (3h)
User management (36h)
User walkthrough (12h)
Biometric data walkthrough (12h)
Watermark management (27h)
watermark management (15h)
Backup management (17h)
Backup walkthrough (12h)
Deployment management (12h)
Deployment walkthrough (6h)
Development (360h)
TBA (360h)
Admin end of the project (40h)
Report (60h)
Today
Intermediary report
May 20
Development focus decided
May 20
Final report
Jul 23

Figure 1: Inital planning for the project

3 State of the Art

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fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus.

4 Repository structure

This chapter focus on the repositories structure as well as their interaction with one another. It is used as a backbone to understand the dependencies of the directories as well as how they're linked together through the CICD.

4.1 List of directories

- afis_assignments
- base
- cdn
- docker
- documents
- fingerprintexperts_assessment
- config/production
- tools
- web

4.2 Description

4.2.1 Repo afis_assignemnts

This repository contains one R file. This file is name `TrialSortingScript` and relies on a csv file that is not present. This seems to be filtering out some data from labs. This seems to be outside the scope of this thesis. It may be removed in the future.

4.2.2 Repo base

This repository contains on Dockerfile as well as a gitlab cicd file and a license. The Dockerfile is a minimal debian install. It updates, upgrades and installs `gettext-base`. It then removes the `/var/lib/apt/lists/*`.
The cicd, when run, uses Kaniko to build the Docker image and pushes it to Gitlab's container registry with two tags. The commit SHA and latest. It has only seen 7 commits and the last one dates from 2022.

There are two images in the container repository of this repo:
- base/cache
- base

4.2.3 Repo cdn

This repo hosts all the js and css that is considered as libraries for the web application. Here's an exhaustive lists of what is present within the repo:
- chosen
- aes
- pbkdf2
- dropzone
- jquery
- loadingcss
- md5.min
- rangeslider.min
- toastr.min
- base64
- moment.min
- otplib-browser
- sha512
- underscore-min

4.2.4 Repo docker

This repo has the description "ICNML application builder process". From the README, we can understand that the docker repo has submodules. After testing, today it cannot reach them. This repository is mainly focused on the CICD pipeline.
From the timestamps of the last commit on the web repository and this one, it seems there's an automation in place to keep this repository updated with the web one.
The README emphasizes that this is not for development and then gives info on how to setup a development environment with the submodules. It is not clear what the responsability or scope of this repository is.
We can see that the web repository is a submodule of this one.
There are 3 images in the container repository of this repo:
- docker/web/cache
- docker/redis
- docker/web

4.2.5 Repo documents

There is only one pdf file within this repository. It is a donor consent form.

4.2.6 Repo fingerprintexperts_assessment

This repo contains the results from tests on ICNML images with fingerprint experts. It's out of scope. It may be deleted later on.

4.2.7 Repo config/production

This repo seems to be a part of an automated process. The timestamps of the last commits correspon to the ones on the web repository.

4.2.8 Repo scripts

This repo hosts scripts used database management. There is one for creating the backup, one for cleaning the backups, one to duplicate the database and one for combining gpg keys.

4.2.9 Repo tools

This repo contains scripts in python to migrate data and convert it. It relies on submodules that are not accesible today.

4.2.10 Repo web

This repo is where the web application code is hosted. It has the cdn repo as submodule, however I also see some dependencies on MDmisc and NIST libraries. There is a flask application, sql schema for a database. There are also a lot of utils files related to encryption, so I think this is where I will spend most of my time after creating a dev environment.

4.3 Schema

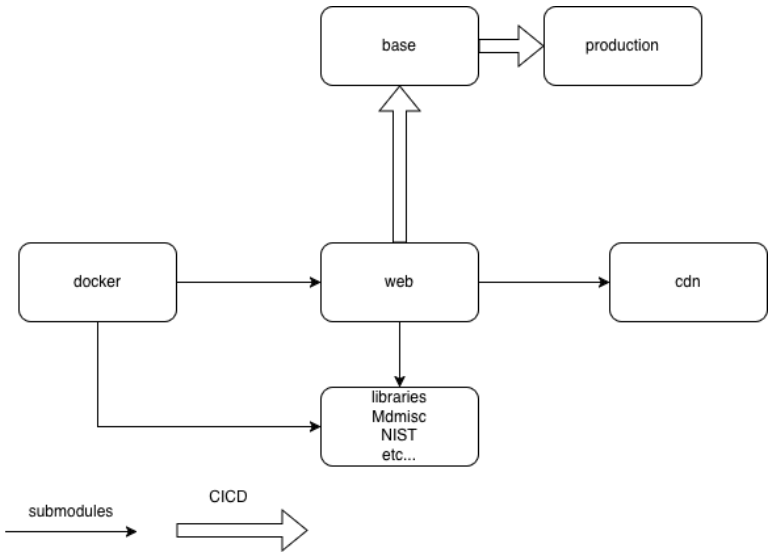

Figure 2: Repositories dependencies

5 Roles and permissions

The application ICNML follows a role-based access control (RBAC) model. Every authenticated user has one and only one role. The different roles possible within the application are stored in the `account_type` sql table. It's reflected on login in the Redis session object. A series of decorator are used to manage the authorisation of each route.
The document describes each role: how they are created, what routes and database operations are allowed, what is explicitely denied.

5.1 Account types

There are 6 account types in the `01-account_type.sql` file. A flag `can_singin` exists for each account type. It filters the types for which a user can request a user account via the `GET /signin` public form.

ID	Name	Can request account	Short description
1	Administrator	No	Full privileged access; manages the platform.
2	Donor	No	Subject whose biometric data is collected; created by a Submitter.
3	Submitter	Yes	Registers donors.
4	Trainer	Yes	Uses mark data to train fingerprint examiners.
5	AFIS	Yes	Works with Automated Fingerprint Identification System targets and candidate matches.
6	Selection	Yes	Selection user, not clear yet what it's useful for.

5.2 Authentication and Session Model

The authentication is handled by the `/login` route. It expects the user to complete a form and then it will call the `POST /do/login` endpoint for each attribute (username, password, TOTP or WebAuthn passkey). Then, these are the keys that are written in Redis.

```python
session[ "account_type" ]
session[ "account_type_name" ]
session[ "logged" ]
session[ "username" ]
session[ "user_id" ]
```

Listing 1: Keys written in Redis' session

5.3 Access-Control decorators

Most of the authorisation logic lies in the decorators file. This is done via the use of `functools` package to make decorator out of functions.

5.3.1 `@login_required`

The first decorator is the one that requires the user to be logged in. It checks that the `session["logged"]` key is set to `True`. Any unauthenticated request is redirected to the login url. This is the baseline access control.

5.3.2 `@admin_required`

This decorator checks wether the `account_type_name` key in the session is set to Administrator. It also checks wether or not the user is logged with the same logic as the `login_required` decorator.
If the the account type name is not Administrator, it will redirect to login.

5.3.3 `@submission_has_access`

This decorator will check wether the user has access to the current submission.

- If the user is an Administrator, then they can access the submission
- If the user is a Submitter, the submission needs to be theirs (this needs clarification, I don't understand it yet)
- All other roles are redirected to the login page

5.3.4 `@trainer_has_access`

This decorator checks wether the `account_type_name` key in the session is set to Administrator. It also checks if this key is set to Trainer and if the user is logged.

If the account type name is not Administrator or Trainer, it will redirect to the login page.

5.4 Role descriptions

5.4.1 Administrator (type 1)

Creation: Accounts must be created directly in the database or by a process not documented.
Home page redirect: `GET /admin/submission/list`
Scope: The Administrator has unrestricted access to the entire application. Every decorator allows Administrators. Dedicated `/admin/*` routes are only usable by them.
Capabilities:

- **New user management**: can validate pending sign in requests via `GET /validate_signin`
- **Submission overview**: can view all submissions via `GET /admin/submission/list`
- **AFIS management**: creates targets (`POST /admin/target/<submission_id>/<pc>/new`), deletes targets or candidate matches, updates assigned AFIS users, and batch-assigns targets (`POST /admin/afis/batch_assign/do`)
- **Full-resolution images:** downloads uncompressed originals via `GET /image/file/<id>/full_resolution`.
- **PiAnoS integration:** synchronises all users and segments to the external PiAnoS system via `GET /pianos_api/add_user/all` and `GET /pianos_api/add_segments/all`.
- **Mark deletion (admin context):** `POST /admin/<submission_id>/mark/<m_id>/delete`.
- **UUID inspection:** retrieves the raw database table for any UUID via `GET /uuid/get_table/<uuid>`.

5.4.2 Donor (type 2)

Creation: Created by a Submitter via `POST /submission/do_new`. No self-registration.
Home page redirect: `GET /user/myprofile/dek`
Scope: Donors manage only their own cryptographic key (DEK) and view their own biometric data. They cannot interact with the submission or AFIS workflows.
Permitted operations:

Route	Purpose
`GET /user/myprofile/dek`	View DEK profile page.
`GET /user/myprofile/tenprint`	View own tenprint cards.
`GET /user/myprofile/marks`	View own mark images.
`POST /dek/reconstruct`	Re-derive the DEK from the donor's e-mail (requires re-authentication).
`GET /dek/delete`	Soft-delete the DEK (`donor_dek.dek` set to `NULL`; reversible).

Route	Purpose
`GET /dek/fulldelete`	Permanently delete the DEK row (right-to-erasure; irreversible).

5.4.3 Submitter (type 3)

Creation: Self-registration via `GET /signin` / `POST /do/signin` followed by e-mail confirmation and administrator approval.
Home page redirect: `GET /submission/list`
Scope: Submitters manage the complete lifecycle of their own submissions, from donor registration through file upload and metadata annotation. They cannot access another submitter's data.
Permitted operations (own submissions only):

Action	Routes
Create donor + submission	`POST /submission/do_new` — creates `users`, `donor_dek`, and `submissions` rows.
Upload files	`POST /upload`, `/submission/<id>/add_files`, `/submission/<id>/add_marks`, `/submission/<id>/consent_form`.
Annotate tenprints	Set template, quality, segment coordinates, and general pattern on tenprint cards.
Annotate marks	Set PFSP?, and delete marks.
Manage submission	Set nickname, set GP, delete submission, view targets.
Browse own data	`GET /submission/list`, tenprint list, mark list, segment views.

Ownership enforcement: `@submission_has_access` issues HTTP 403 if the `submission_id` in the URL was not created by the current submitter.

5.5 Trainer (type 4)

Creation: Self-registration via `GET /signin`.
Home page redirect: `GET /marks/search`
Scope: Trainers consume mark and tenprint data for examiner-training exercises. Their access is read-oriented; they do not create submissions or manage donors.
Permitted operations:

Route	Purpose
`GET /marks/search`	Search and filter marks for training.
`GET /marks/exercise/<id>`	View a training exercise.

Route	Purpose
`GET /marks/folder/<id>`	Browse an exercise folder.
`POST /exercises/add_tenprint`	Associate a tenprint card with an exercise.

5.5.1 AFIS (type 5)

Creation: Self-registration via `GET /signin`.
Home page redirect: `GET /afis/list/targets`
Scope: AFIS users work on fingerprint identification targets: they receive candidate match assignments, upload and annotate search results, and set comparison decisions. This seems like a big part of the application.
Permitted operations:

Route	Purpose
`GET /afis/list/targets`	List assigned AFIS targets.
`GET /afis/incidental/donors/list`	List incidental donors.
`GET /afis/incidental/donor/<uuid>/list`	Browse a specific incidental donor's data.
`GET /afis/<uuid>`	View an AFIS target.
`GET /afis/<uuid>/download{,/mark,_exercise}`	Download target data, mark, or exercise package.
`GET /afis/<uuid>/upload/list`	List uploaded result files.
`GET /afis/<uuid>/upload/new/<type>`	Initiate a result file upload.
`GET /afis/<target>/<cnm>`	View a candidate match.
`POST /afis/<target>/<cnm>/set_pfsp`	Record the PFSP decision for a candidate match.
`POST /afis/<target>/<cnm>/upload`	Upload a candidate match result file.
`POST /afis/<target>/<cnm>/update_field`	Update a field on a candidate match.
`GET /afis/<cnm>/<file>/<fpc>/autodetect`	Auto-detect minutiae from a mark file.
`GET /afis/<cnm>/<file>/autodetect/tiff`	Auto-detect from a TIFF file.
`GET /afis/<cnm>/<file>/<fpc>/res`	Retrieve image resolution metadata.

Route	Purpose
`GET /image/cnm_candidate/screenshot/<file>/preview`	Preview a candidate screenshot.

5.5.2 Selection (type 6)

Creation: Self-registration via `GET /signin`.
Home page redirect: Default (`/`) — no specialised redirect is defined in `views/base/__init__.py` for this role.
Scope: The Selection role is defined in the database and the registration flow but has no dedicated routes or decorators at this time. A Selection user can log in and reach any route guarded only by `@login_required` (shared utilities, UUID search, image preview, etc.), but cannot access submission, AFIS, trainer, or admin-specific routes.

6 DEK Donor Generation

When a new donor is registered within the ICNML application, the first cryptographic operation is the generation of a **Data Encryption Key** (DEK). This key is unique to the donor and is used to encrypt all biometric data associated with them.
The DEK is designed to be completely **uncoupled** with the rest of the application so that deleting it will render all biometric data within the database unrecoverable. This is a very important workflow to guarantee a kind of privacy-by-deletion design. If the user wants to delete their key, they have the possibility to do it, and it will render all their biometric data unreadable by other users.
This document describes the complete DEK generation workflow and is used as a stepping stone for future documentation. This will cover the HTTP routes used, the cryptographic steps, and the object stored in the database to fully understand the process.

6.1 Prerequisites and Roles

The donor creation process, by derivation the DEK generation, requires a **Submitter** and the request from the user to be a **Donor**.

6.2 Steps

6.2.1 Entry point

The entry point for the generation of the donor dek is `POST /submission/do_new`. This is triggered after the **Submitter** has sent the form to register the new **Donor**. This is only accessible for the users with **Administrator** or **Submitter** role.
The form provides two inputs:

- `email`, this is the donor's plain-text email
- `upload_nickname`, this is the submission nickname

6.2.2 Step 1, Duplicate Check

Before any creation, the code checks wether the email is already used within the db. To be more precise from the sql query, it will check if, among the submissions done by the **Submitter**, one has the same email hash.

```python
sql = "SELECT id, email_hash FROM submissions WHERE submitter_id = %s"
...
if utils.hash.pbkdf2( email ).verify( case[ "email_hash" ] ):
# Return error that email is used for another submission
```

Listing 2: Code for duplicate check

We can see here that the provided email is hashed and then is verified against the `email_hash` stored in the database. If a match is found the request is rejected with an error.

6.2.3 Step 2, Donor User Creation Account

In this step, the **Donor** is created with the status *pending*. The `email` and the `upload_nickname` are both encrypted using the aes functions found in `utils/aes.py`. Then an id is retrieved via a sequence found in the database.
A new user is inserted within the table `users` with the format `donor_<id>`. Here is stored the username, the email as well as the type.

```python
    userid = config.db.query_fetchone( "SELECT nextval( 'username_donor_seq' ) as id" )[ "id" ]
    username = "donor_{}".format( userid )
    sql = utils.sql.sql_insert_generate( "users", [ "username", "email", "type" ], "id" )
```

Listing 3: Creation of a new donor

6.2.4 Step 3, DEK Generation

After creating a user of type **Donor**, the DEK is generated calling the `utils.encryption.dek_generate()` function. This function can take up to 4 arguments:

- `email`, the email of the donor
- `email_hash`, the hash of the donor's email
  - Either `email` or `email_hash` must be present in the kwargs
- `username`, the username of the donor (required)
- `salt`, the salt used for key generation (optional)

The function will hash the username and the email hash given using pbkdf2 to create the dek.

```python
    dek = hash.pbkdf2(
        "{}:{}".format( username, email, ),
        dek_salt,
```

```
        iterations = config.DEK_NB_ITERATIONS,
        hash_name = "sha512"
    ).hash( True )
```

Listing 4: Code for the DEK generation

The `config.DEK_NB_ITERATIONS` is 500′000.
The function creates a check object with a call to `aes.do_encrypt()` with a json as parameters and the dek. It is used later on if the user chooses to recreate their DEK after a soft-delete.
The function returns 3 variables:

- `dek_salt`, the salt used for generating the DEK
- `dek`, the generated DEK
- `check`, the encrypted DEK for rebuilding the DEK later on to check against

6.2.5 Step 4, DEK persistance

After the creation of the DEK, it is stored within the `donor_dek` table. Here's a list of the different fields that are persisted:

- `donor_name`, this is the username created previously `donor_<id>`
- `salt`, this is the `dek_salt` returned by the dek generating function
- `dek`, the generated DEK
- `dek_check`, the encrypted DEK using AES
- `iterations`, the number of iterations used to create the DEK
- `algo`, the algo used for generating the DEK, here always pbkdf2
- `hash`, the structure of the hash used, here always sha512

It then returns a response with error false and the donor uuid.

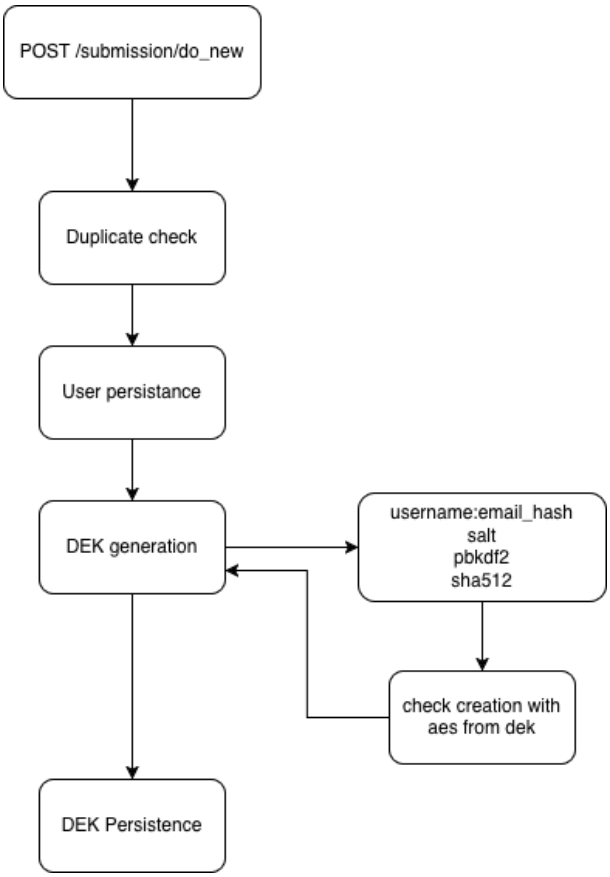

Figure 3: DEK Generation Schema

7 Architecture

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8 Implementation

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9 Results

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10 Conclusion

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I Tools used

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II Work diary

Date	Description	Rech. [h]	Dev. [h]	Rapport [h]	Admin [h]
					Le journal de travail continue à la page suivante.
05.03.2026	Kick-off meeting avec le mandant	0	0	0	2
12.03.2026	Recherche sur l'état de l'art	4	0	0	0
19.03.2026	Mise en place de l'environnement de développement et documentation	0	7	1	0
25.03.2026	Rédaction de l'état de l'art	0	0	4	0
25.03.2026	Dev de la partie interface	0	0	4	0
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0

Date	Description	Rech. [h]	Dev. [h]	Rapport [h]	Admin [h]
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0
xx.xx.2026	Test	0	0	4	0

Table 2: Journal de travail